

Sohag Kumar Saha

☎ +1 (931) 252 9504 ✉ ssaha42@tntech.edu  [linkedin.com/in/sohag16](https://www.linkedin.com/in/sohag16)  sohagkumarsaha.github.io

CAREER SUMMARY

Power Systems Engineer with 10+ years of experience spanning grid operation and maintenance, utility-scale substation construction, and transmission & distribution network modeling. Proficient in end-to-end project execution, large-scale power flow analysis using OpenDSS-Python, and real-time control validation in Typhoon HIL. Strong background in evaluating power system matrices, benchmarking simulation performance, and bridging the gap between hardware execution and high-performance computational grid studies, coupled with hands-on experience in designing BESS plant controls, inverter-based resources, HIL SCADA integration, and utility communications (Modbus TCP/IP, IEC 61850). Demonstrated ability to interface effectively with cross-functional teams, utilities, EPC contractors, and internal stakeholders to deliver reliable and compliant power system projects. Passionate about deploying cutting-edge energy technologies to accelerate the transition to a resilient and sustainable grid.

Employment Authorization: Authorized to work in the U.S. as a Permanent Resident (PR) – no sponsorship required.

EDUCATION

Ph.D. in Electrical and Computer Engineering (ECE) Aug. 2021–Present (Expected Graduation: TBD)
Tennessee Technological University, Cookeville, TN, USA.

Research Focus: Power Systems Modeling, Energy Management Systems, and CHIL Validation.

M.Sc. in Electrical and Computer Engineering (ECE) Aug. 2021-Dec. 2024
Tennessee Technological University, Cookeville, TN, USA.

B.Sc. in Electrical and Electronic Engineering (EEE) Mar. 2009-Dec. 2013
Pabna University of Science and Technology, Pabna, Bangladesh.

RELEVANT COURSEWORKS

- Renewable Energy, Switchgear and Protection, Power Plant Engg., Power and Industrial Electronics, Electrical Power Trans. and Distribution, Power System Engg., Computer Methods of Power System Analysis, Electromagnetic Fields and Theory
- Control System Engg., Linear Control System, Non-linear Auto. Control, Learning and Adaptive Control, Machine Learning, Cyber-Physical Security, Internet Security, Electrical Measurement and Instru., Signal Analysis, Wireless Power Transfer

SKILLS SUMMARY

- **Programming Languages:** MATLAB, Python, C, C++, and Structured Text
- **Library+Frameworks:** Matplotlib, Pandas, Seaborn, Scikit-learn, TensorFlow, PyTorch, Keras
- **Communication Protocols:** Modbus RTU, Modbus TCP/IP, IEC-61850, Ethernet
- **Software Tools:** Simulink, Typhoon HIL, OpenDSS, PSCAD, LTspice, Proteus, Multisim, OpenPLC, SCADAbr
- **Hardware Tools:** Typhoon HIL, Universal HIL Connect, SEL RTAC, EPC Power Inverter Controller, SEL Protection Relays
- **Platforms:** Windows, Kali Linux, Arduino IDE
- **Typesetting:** Microsoft Word, Overleaf LaTeX, Mendeley Citation Tool
- **Substation Construction:** Physical Layouts; Bus-bar Arrangement; Grounding Design; Lightning Protection; DERs Sizing
- **Switchgear Equipment Testing:** 230kV/132kV/33kV AIS & 400 kV GIS Switchgear; SF6 Circuit Breakers; Power Transformers, Auto-transformers; Instrument Transformers (CT/PT); Surge Arresters; Disconnect Switches; and Protection Relays.
- **Project Management:** Bill of Materials Management; Vendor Specification Preparation and Review; Technical Change Control; Risk Management; Contractor Supervision; Commissioning & Energization Plans; Keep Records and Design Updates.
- **Power System Modeling, Analysis and Optimization:** Distribution network modeling, Power flow, DC-OPF, AC-OPF, State estimation, Power system optimization, Convex Optimization, PSO Optimization, Deep Learning and Reinforcement Learning

PROFESSIONAL EXPERIENCES

Graduate Research Assistant (GRA) — ECE, Tennessee Tech University (TTU) TN, USA
Center for Energy Systems Research (CESR) Aug. 2021 - Till Date

- Intelligent Energy Management Systems within Community Microgrid using Real-time Controller Hardware-In-the-Loop.
- Developed and validated an intelligent site-control and energy management system for a community microgrid using real-time controller-HIL in Typhoon, SEL RTAC, and EPC power inverter controllers, implementing sequence of operations for grid-connected and islanded operations with actor-critic-based adaptive critic design.
- Implemented supervisory control logic on SEL RTAC 3530-4 using PLC Structured Text (AcSElerator RTAC) to generate and regulate active/reactive power setpoints for BESS/PV/wind/microgrid assets designed in Typhoon HIL.

- Developed Modbus TCP/IP integrated BESS plant operations for peak-shaving applications; verified control behavior in HIL.
- EV charging control and SoC balancing using consensus-based tracking control in Typhoon HIL, RTAC and inverter controller.
- Integrated IEC-61850 GOOSE communications to evaluate grid communication behavior relevant to interoperable site control.
- Developed and evaluated control strategies for DAB converter control using Simulink and Typhoon HIL-based verification.
- Performed distribution-system time-series studies using OpenDSS, including PV and EV hosting capacity analysis, and analyzed high-resolution simulation data using Python to implement optimization for dispatch/EMS studies for economic load dispatch with adaptive particle swarm optimization evaluating performance and optimizing computational runtimes.
- Documented modeling assumptions, validation results, and technical findings for presentations and cross-functional teams.

Sub-Divisional Engineer/AE — Ashuganj Power Station Company Ltd. (APSCL) Brahmanbaria, Bangladesh
Sub-station Construction, Operation and Maintenance Division Aug. 2016 - Aug. 2021

- Led LV and HV electrical installation, inspection, testing, commissioning, and maintenance of HV grid substation equipment (33 kV, 132 kV, & 230 kV), coordinating commissioning activities and resolving issues during energization and operations.
- Managed the project bill of materials (BOM) and reviewed technical specifications for high-voltage equipment procurement, supporting sourcing teams with vendor quotation reviews and technical evaluations.
- Directed a cross-functional team of engineers and technicians during scheduled outages and emergency maintenance, ensuring adherence to safety protocols and minimizing downtime.
- Conducted root cause analysis on switchgear and protection system faults; implemented corrective design changes to improve grid availability and reliability.
- Executed preventive maintenance and served as technical contact for external EPC contractors during grid expansion projects.
- Grid commissioning highlights: installation, testing, and commissioning of HV substation assets, including autotransformers, interbus coupling transformers, SF6 breakers, shunt reactors, and multiple distribution transformers and switching stations.

Assistant Engineer (AE) — Bangladesh Rural Electrification Board (BREB) Dhaka, Bangladesh
Power Distribution Network Operation, Control and Maintenance Aug. 2015 - Aug. 2016

- Supervised the construction and commissioning of 33/11 kV distribution substations, including physical layout inspection, civil works inspection, grounding mesh installation, and control building wiring.
- Supported the operation of distribution networks to ensure power quality and reliability for rural electrification projects.
- Managed progress reporting and technical documentation for ongoing grid connection projects, coordinating between field teams and the controlling authority.

POSTER PRESENTATIONS

- **Sohag Kumar Saha**, Ali Arzani and Satish M. Mahajan, “Intelligent Energy Management System for Community Microgrids.” *Tennessee Renewable Energy & Economic Development Council (TREEDC) conference*, Leslie Town Center, Cookeville, TN, USA, 13 Nov 2025.
- **Sohag Kumar Saha**, Ali Arzani and Satish M. Mahajan, “HILLTOP-Based BESS Plant for Peak-Shaving Application.” *Tennessee Renewable Energy & Economic Development Council (TREEDC) conference*, Leslie Town Center, Cookeville, TN, USA, 14 Nov 2024.
- **Sohag Kumar Saha** and Douglas A. Talbert, “Automatic Detection of Dust Deposition in Solar Photovoltaic Panels using CNN and LSTM Algorithms.” *Research and Creative Inquiry Day-2023*, Tennessee Tech University, Cookeville, TN, USA, 19 April 2023.

PUBLICATIONS

- **Sohag Kumar Saha**, Ali Arzani and Satish M. Mahajan, “Meta-DQN Tuned TD3-based Safety-Aware Deep Reinforcement Learning for Energy Management System in Community Microgrids.” *Accepted in the IEEE Industry Applications Society (IAS) Annual Meeting 2026 Conf., Vancouver, Canada*. [Accepted]
- Hany A. Abdelsalam, Ali Arzani, Satish M. Mahajan, and **Sohag Kumar Saha**, “Real-Time Implementation of Consensus Tracking Control for Battery Energy Management and SoC Balancing.” *in Proceedings of the IEEE 10th Texas Power and Energy Conference (TPEC)*, College Station, Texas, Feb. 2026, pp. 1–6.
- Prashant Kumar, Ali Arzani, **Sohag Kumar Saha**, and Satish M. Mahajan, “An Enhanced Linear Active Disturbance Rejection Controller-Based Dual Active Bridge Converters for EV Charging.” *in Proceedings of the IEEE Energy Conversion Conference and Expo (ECCE)*, Philadelphia, Pennsylvania, USA, October 2025, pp. 1–6.
- **Sohag Kumar Saha**, Ali Arzani, Reynaldo Salcedo, and Satish M. Mahajan, “Modbus TCP/IP Based BESS Plant Controller Operations for a Peak Shaving Application.” *in Proceedings IEEE 9th Texas Power and Energy Conference (TPEC)*, College Station, Texas, USA, February 2025, pp. 1–6.

- **Sohag Kumar Saha** and Satish M. Mahajan, “Multivariate Optimal Hybrid Deep Learning Model for Forecasting of Day-Ahead Solar Irradiance with Meteorological Constraints.” in *Proceedings of IEEE 56th North American Power Symposiums (NAPS)*, El Paso, Texas, USA, 2024, pp. 1-6.
- Shampa Banik, **Sohag Kumar Saha**, Trapa Banik and S M Mostaq Hossain, “Anomaly Detection Techniques in Smart Grid Systems: A Review.” in *Proceedings of IEEE World AI IoT Congress (AIIoT)*, Seattle, USA, May 2023, pp. 1-7.
- Shampa Banik, Trapa Banik, S M Mostaq Hossain, **Sohag Kumar Saha**, “Implementing Man-In-The-Middle Attack to Investigate Cyber-Security Vulnerabilities in Smart-Grid Test-Bed.” in *Proceedings of IEEE World AI IoT Congress (AIIoT)*, Seattle, USA, May 2023, pp. 1-7.
- S M Mostaq Hossain, **Sohag Kumar Saha**, Shampa Banik and Trapa Banik, “A New Era of Mobility: Exploring Digital Twin Applications in Autonomous Vehicular Systems.” in *Proceedings of IEEE World AI IoT Congress (AIIoT)*, Seattle, USA, May 2023, pp. 1-7.
- **Sohag Kumar Saha** and Md. Selim Hossain, “Novel Approach of Antenna Array with Beam Steering Technology for Microwave Power Transmission from Space Solar Power Satellite (SSPS) System.” in *Proceedings of 2nd Int’l Conference on Electrical, Computer & Telecommunication Engineering (ICECTE)*, Rajshahi, Bangladesh, December 2016, pp. 1-4.

ACADEMIC HONORS AND AWARDS

- Ongoing Ph.D. research funded by the U.S. Appalachian Regional Commission (ARC) under Grant MU-21579-23.
- Recipient of multiple travel grants, including the NSF Grant (TPEC 2025, College Station, TX) and CESR Travel Grants for NAPS 2024 (El Paso, TX) and Appalachian Energy Summit 2023 (Boone, NC), supporting oral research presentations.
- Bangladesh-Sweden Trust Fund Scholarship 2022 as a travel grant for covering Bangladesh to USA flight to conduct Ph.D.
- BASIS Outsourcing Award as Best Freelancer of the Year-2014 in Web Development Category (District-wise award in BD).

TECHNICAL MEMBERSHIP

- **Senior Member, IEEE**; Region: R3 -Southeastern USA, Section: Central Tennessee, USA.
- **Senior Member**, IEEE Power and Energy Society (IEEE-PES), IEEE Power Electronics Society (IEEE-PELS), and IEEE Industry Applications Society (IEEE-IAS).
- **Member**, International Council on Large Electric Systems (CIGRE), Type: Individual Member-III.
- **Member**, Bangladesh Water and Power Engineers Association (BWPEA).

VOLUNTARY PEER REVIEWER

- **Journals:** IEEE Transactions on Transportation Electrification; IEEE Transactions on Power Electronics; IEEE Access; DE Gruyter: International Journal of Emerging Electric Power Systems; Springer: Peer-to-Peer Networking and Applications; International Journal of Renewable Energy Resources; Indonesian Journal of Electrical Engineering and Computer Sciences.
- **Conferences:** IEEE PES General Meeting; IEEE PES International Meeting (2026, Hong Kong SAR, China); IEEE North American Power Symposium (NAPS) 2026, IEEE Industry Application Society Annual Meeting (IASAM 2026), IEEE Applied Power Electronics Conference and Exposition (APEC) 2026, IAS Annual Meeting 2026, IEEE ECCE 2025 (PA, USA); IEEE ECCE Europe 2025 (Birmingham, UK); IEEE ITEC 2025 & 2026; IEEE TPEC 2025 (College Station, TX, USA); IEEE PES Innovative Smart Grid Technologies 2024; IEEE PES/IAS PowerAfrica Conference; IEEE SmartGridComm; IEEE COBEP-SPEC; Int’l Conference on Artificial Intelligence, Computer, Data Sciences and Applications; International Conference on Electrical, Computer and Energy Technologies; Int’l Conference on Smart and Sustainable Developments in Electrical Engineering; Int’l Conference on Electrical, Computer, Communications and Mechatronics Engineering.

SELECTED TECHNICAL TRAINING AND PROFESSIONAL DEVELOPMENT

- Completed specialized training in **Typhoon HIL**, including HIL Fundamentals, HIL Specialist-2, Grid-Forming Inverters, and Distributed Secondary Control for Microgrids (Typhoon HIL Academy, UIUC, Federal University of Santa Maria).
- Completed professional training on **Power Systems and Smart Grid Technologies**, including DER management using SEL RTAC Grid Connect, Smart Grid modernization (Power-China, China), power system reliability, transformer operation & maintenance, and protection, metering, and control of high-voltage grid systems.
- Participated in the **Nashville Electric Service (NES)** Control Room and Substation industrial visit, including a live demonstration of the SEL Microgrid system (Nashville, TN, USA).
- Completed professional development courses in **Python, Data Science, MATLAB**, ISO 9001:2015, ISO 14001:2015, ISO 45001:2018, Lab Safety, and Instructor Certification (iLearn, Tennessee Tech).